

BRUFEN® 100MG/5ML (IBUPROFEN)

PRODUCT DESCRIPTION

Brufen® Suspension 100mg/5ml contains ibuprofen 100mg per 5ml as active ingredient, available as an orange coloured and flavoured syrup suspension.

Ibuprofen is a member of the propionic acid group of nonsteroidal anti-inflammatory drugs (NSAIDs). Ibuprofen is a racemic mixture of [+] S- and [-] R-enantiomers.

Ibuprofen has shown anti-inflammatory analgesic and antipyretic activity in both animal and human studies. These properties provide symptomatic relief of inflammation and pain.

INDICATIONS

Ibuprofen is indicated for the treatment of rheumatoid arthritis, including juvenile arthritis or Still's disease, ankylosing spondylitis, osteoarthritis and other non-rheumatoid (seronegative) arthropathies and acute gouty arthritis. It is also indicated for the treatment of non-articular rheumatic conditions and periarticular conditions such as frozen shoulder (capsulitis), bursitis, tendinitis, tenosynovitis and low back-pain.

It is also used in the treatment of soft-tissue injuries such as sprains and strains, for relief of mild to moderate pain such as primary dysmenorrhea, dental and post-operative pain, post-episiotomy pain, post-partum pain and for the symptomatic relief of headache including migraine headache.

In addition, the suspension is indicated in short-term use for the treatment of pyrexia in children over one year of age.

DOSAGE AND ADMINISTRATION

Adults

The recommended dose is 1200 to 1800 mg/day in divided doses. Some patients can be maintained on 600 to 1200 mg/day. In severe and acute cases, it could be advantageous to increase the dosage until the acute phase is over. Daily dose should not exceed 2400 mg in divided doses. However, the dose could be raised to 3200 mg if it is necessary but the patient has to be monitored closely.

The recommended daily dosage of slow release ibuprofen is two tablets as a single dose preferably in the early evening, well before retiring to bed. The tablets should be swallowed whole with plenty of fluid. In severe or acute conditions, the total daily dose may be increased to three tablets in divided doses.

After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

Elderly

No special dose modifications are necessary unless renal or hepatic function is impaired to which case dosage should be assessed individually. Caution should be taken with dosage in this group.

In all indications, the dose should be adjusted for each patient and the smallest dosed employed.

Children

Daily dose of ibuprofen is 20mg/kg body weight in divided doses. This can be achieved by using the syrup as follows.

AGE	FREQUENCY	DOSE
1 to 2 years	2.5 mL (50 mg)	3 to 4 times a day
3 to 7 years	5 mL (100 mg)	3 to 4 times a day
8 to 12 years	10 mL (200 mg)	3 to 4 times a day

Ibuprofen is not recommended for children weighing less than 7 kilograms. Ibuprofen 800, slow release Ibuprofen and Ibuprofen granules are not suitable for children under the age of 12 years. In juvenile rheumatoid arthritis, up to 40 mg/kg body weight per day in divided doses may be taken.

CONTRAINDICATIONS

- Known hypersensitivity to the active substance or to any of the inactive ingredients.
- Ibuprofen should not be given to patients who have experienced asthma, urticaria or allergic-type reactions after taking acetylsalicylic acid/aspirin or other NSAIDs.
- Severe heart failure.
- Severe liver failure.
- Severe renal failure (glomerular filtration below 30 mL/min).
- Conditions involving an increased tendency to bleeding.
- History of gastrointestinal bleeding or perforation, related to previous NSAID therapy.
- Active, or a history of, ulcerative colitis, Crohn's disease, recurrent peptic ulceration or gastrointestinal hemorrhage (defined as two or more distinct episodes of proven ulceration or bleeding).
- During the third trimester of pregnancy.

WARNINGS AND PRECAUTIONS

General

Undesirable effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see section Dosage and Administration).

On prolonged use of any painkillers, headache may occur that must not be treated with increased doses of the medicinal product.

Through concomitant consumption of alcohol, active substance-related undesirable effects, particularly those that concern the gastrointestinal tract or the central nervous system, may be increased on use of NSAIDs.

Ibuprofen 20 mg/mL oral suspension contains sucrose and sorbitol. Patients with rare hereditary problems of fructose intolerance, glucose-galactose malabsorption or sucrose-isomaltase insufficiency should not take this medicine. This should also be taken into account in patients with diabetes mellitus. May be harmful to the teeth. Ibuprofen 20 mg/mL oral suspension contains methyl parahydroxybenzoate and propyl parahydroxybenzoate. May cause allergic reactions (possibly delayed). Ibuprofen 20 mg/mL oral suspension contains sunset yellow (E110). May cause allergic reactions.

Elderly population

Elderly patients have an increased frequency of adverse reactions to NSAIDs, especially gastrointestinal bleeding and perforation, which may be fatal.

- **Gastrointestinal Bleeding, Ulceration and Perforation**

Ibuprofen should be given with care to patients with a history of peptic ulceration and other gastrointestinal disease since their conditions may be exacerbated (see section Contraindications).

Gastrointestinal bleeding, ulceration or perforation has been reported with all NSAIDs at any time during treatment. These adverse events can be fatal and may occur with or without warning symptoms or a previous history of serious gastrointestinal events.

The risk of gastrointestinal bleeding, ulceration or perforation is higher with increasing ibuprofen doses in patients with a history of ulcers, particularly if complicated with hemorrhage or perforation, and in the elderly. These patients should commence treatment on the lowest dose available.

Combination therapy with protective agents (e.g. misoprostol or proton pump inhibitors) should be considered for these patients, as well as patients requiring concomitant low dose acetylsalicylic acid/aspirin, or for other drugs likely to increase gastrointestinal risk (see section Drug Interactions).

The concomitant administration of ibuprofen and other NSAIDs, including cyclooxygenase-2 (Cox-2) selective inhibitors, should be avoided due to the increased risk of ulceration or bleeding (see section Drug Interactions).

Patients with a history of gastrointestinal disease, particularly when elderly, should report any unusual abdominal symptoms (especially gastrointestinal bleeding) in the initial stages of treatment.

Caution should be exercised in patients receiving concomitant medication which could increase the risk of ulceration or bleeding, such as oral corticosteroids, anticoagulants such as warfarin, selective serotonin re-uptake inhibitors or antiplatelet drugs such as acetylsalicylic acid/aspirin (see section Drug Interactions).

If gastrointestinal bleeding or ulceration occurs in patients receiving ibuprofen, the treatment should be withdrawn.

- **Respiratory Disorders**

Caution is required if ibuprofen is administered to patients suffering from, or with a previous history of bronchial asthma, chronic rhinitis or allergic diseases since ibuprofen has been reported to cause bronchospasm, urticaria or angioedema in such patients.

- **Cardiac, Renal and Hepatic Impairment**

Caution is required in patients with renal, hepatic or cardiac impairment since the use of NSAIDs may result in deterioration of renal function. The habitual concomitant intake of similar painkillers further increases this risk. For patients with renal, hepatic or cardiac impairment, use the lowest effective dose, for the shortest possible duration and monitor renal function especially in long-term treated patients (see section Contraindications).

- **Cardiovascular and Cerebrovascular Effects**

Ibuprofen should be given with care to patients with a history of heart failure or hypertension since edema has been reported in association with ibuprofen administration.

Clinical studies suggest that use of ibuprofen, particularly at a high dose (2400 mg daily) may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke). Overall, epidemiological studies do not suggest that low dose ibuprofen (e.g., ≤1200 mg daily) is associated with an increased risk of arterial thrombotic events.

Patients with uncontrolled hypertension, congestive heart failure (NYHA II-III), established ischemic heart disease, peripheral arterial disease, and/or cerebrovascular disease should only be treated with ibuprofen after careful consideration and high doses (2400mg/day) should be avoided.

Careful consideration should also be exercised before initiating long-term treatment of patients with risk factors for cardiovascular events (e.g., hypertension, hyperlipidemia, diabetes mellitus, smoking), particularly if high doses of ibuprofen (2400 mg/day) are required.

Cases of Kounis syndrome have been reported in patients treated with Ibuprofen. Kounis syndrome has been defined as cardiovascular symptoms secondary to an allergic or hypersensitive reaction associated with constriction of coronary arteries and potentially leading to myocardial infarction.

- **Dermatological Effects**

Severe cutaneous adverse reactions (SCARs)

Severe cutaneous adverse reactions (SCARs), including exfoliative dermatitis, erythema multiforme, Stevens-Johnson syndrome (SJS) Toxic Epidermal Necrolysis (TEN), Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS syndrome) and acute generalized exanthematous pustulosis (AGEP), which can be life-threatening or fatal, have been reported in association with the use of Ibuprofen (see section Adverse Reaction). Most of these reactions occurred within the first month. If signs and symptoms suggestive of these reactions appear ibuprofen should be withdrawn immediately and an alternative treatment considered (as appropriate). Exceptionally, varicella can be at the origin of serious cutaneous and soft tissues infectious complications. To date, the contributing role of NSAIDs in the worsening of these infections cannot be ruled out. Thus, it is advisable to avoid use of Ibuprofen in case of varicella.

- **Renal Effects**

Caution should be used when initiating treatment with ibuprofen in patients with considerable dehydration. There is a risk of renal impairment especially in dehydrated children adolescents and the elderly.

As with other NSAIDs, long-term administration of ibuprofen has resulted in renal papillary necrosis and other renal pathologic changes. Renal toxicity has also been seen in patients in whom renal prostaglandins have a compensatory role in the maintenance of renal perfusion. In these patients, administration of a NSAIDs may cause a dose-dependent reduction in prostaglandin formation and secondarily, in renal blood flow which may cause renal failure. Patients at greatest risk of this reaction are those with impaired renal function, heart failure, liver dysfunction, those taking diuretics and ACE inhibitors and the elderly. Discontinuation of nonsteroidal anti-inflammatory drug therapy is usually followed by recovery to the pretreatment state.

Renal tubular acidosis and hypokalaemia may occur following acute overdose and in patients taking ibuprofen products over long periods at high doses (typically greater than 4 weeks), including doses exceeding the recommended daily dose.

- **Hematological Effects**

Ibuprofen like other NSAIDs can inhibit platelet aggregation and prolong bleeding time in normal subjects.

- **Aseptic Meningitis**

Aseptic meningitis has been observed on rare occasions in patients on ibuprofen therapy. Although it is probably more likely to occur in patients with systemic lupus erythematosus and related connective tissue diseases. It has been reported in patients who do not have an underlying chronic disease.

- **Masking of symptoms of underlying infections**

As with other NSAIDs, Ibuprofen can mask symptoms of infection, which may lead to delayed initiation of appropriate treatment and thereby worsening the outcome of the infection. This has been observed in bacterial community acquired pneumonia and bacterial complications to varicella. When ibuprofen is administered for fever or pain relief in relation to infection, monitoring of infection is advised. In non-hospital settings, the patient should consult a doctor if symptoms persist or worsen.

DRUG INTERACTIONS

Care should be taken in patients treated with any of the following drugs as interactions have been reported in some patients:

Concomitant use of ibuprofen with:	Possible effects:
Other NSAIDs including cyclooxygenase-2 selective inhibitors	Concomitant use with other NSAIDs, including cyclooxygenase-2 selective inhibitors, should be avoided due to the potential for additive effects (see section Warnings and Precautions).
Cardiac glycosides	NSAIDs may exacerbate heart failure, reduce filtration rate and increase plasma levels of cardiac glycosides.
Corticosteroids	Increased risk of gastrointestinal ulceration or bleeding with NSAIDs.
Anticoagulants	NSAIDs may enhance the effects of anticoagulants, such as warfarin.
Antiplatelet-agents & Selective serotonin-reuptake inhibitors (SSRIs) (e.g., clopidogrel and ticlopidine)	Increased risk of gastrointestinal bleeding with NSAIDs
Acetylsalicylic acid/aspirin	As with other products containing NSAIDs, concomitant administration of ibuprofen and acetylsalicylic acid/aspirin is not generally recommended because of the potential of increased adverse effects. Experimental data suggest that ibuprofen may competitively inhibit the effect of low dose acetylsalicylic acid/aspirin on platelet aggregation when they are dosed concomitantly. Although there are uncertainties regarding extrapolation of these data to the clinical situation, the possibility that regular, long-term use of ibuprofen may reduce the cardioprotective effect of low-dose acetylsalicylic acid cannot be excluded. No clinically relevant effect is considered to be likely for occasional ibuprofen use (see section Pharmacodynamics).
Lithium	NSAIDs may decrease elimination of lithium.
Anti-hypertensive, beta-blockers and diuretics	NSAIDs may reduce the effect of anti-hypertensives, such as ACE inhibitors, angiotensin-II receptor antagonists, betablockers and diuretics. Diuretics can also increase the risk of nephrotoxicity of NSAIDs.

Methotrexate	NSAIDs may inhibit the tubular secretion of methotrexate and reduce clearance of methotrexate.
Cyclosporine	Increased risk of nephrotoxicity with NSAIDs.
Tacrolimus	Possible increased risk of nephrotoxicity when NSAIDs are given with tacrolimus.
Zidovudine	Increased risk of hematological toxicity when NSAIDs are given with zidovudine. There is evidence of an increased risk of hemarthrosis and hematoma in HIV(+) hemophiliacs receiving concurrent treatment with zidovudine and ibuprofen.
Quinolone antibiotics	Animal data indicate that NSAID's can increase the risk of convulsions associated with quinolone antibiotics. Patients taking NSAID's and quinolones may have an increased risk of developing convulsions.
CYP2C9 inhibitors	Concomitant administration of ibuprofen with CYP2C9 inhibitors may increase the exposure to ibuprofen (CYP2C9 substrate). In a study with voriconazole and fluconazole (CYP2C9 inhibitors), an increased S(+)-ibuprofen exposure by approximately 80 to 100% has been shown. Reduction of the ibuprofen dose should be considered when potent CYP2C9 inhibitors are administered concomitantly, particularly when high-dose ibuprofen is administered with either voriconazole or fluconazole.
Sulfonylureas	NSAIDs may potentiate the effects of sulfonylurea medications. There have been rare reports of hypoglycemia in patients on sulfonylurea medications receiving ibuprofen.
Cholestyramine	The concomitant administration of ibuprofen and cholestyramine may reduce the absorption of ibuprofen in the gastrointestinal tract. However, the clinical significance is unknown.
Aminoglycosides	NSAIDs may decrease the excretion of aminoglycosides.
Herbal extracts	Ginkgo biloba may potentiate the risk of bleeding with NSAIDs.
Mifepristone	A decrease in the efficacy of the medicinal product can theoretically occur due to the antiprostaglandin properties of NSAIDs including acetylsalicylic acid. Limited evidence suggests that coadministration of NSAIDs on the day of prostaglandin administration does not adversely influence the effects of mifepristone or the prostaglandin on cervical ripening or uterine contractility and does not reduce the clinical efficacy of medicinal termination of pregnancy.

FERTILITY, PREGNANCY AND LACTATION

Fertility

The use of ibuprofen may impair female fertility and is not recommended in women attempting to conceive. In women who have difficulties conceiving or who are undergoing investigation of infertility,

withdrawal of ibuprofen should be considered.

Pregnancy

Inhibition of prostaglandin synthesis may adversely affect the pregnancy and/or embryo/fetal development. Data from epidemiological studies suggest an increased risk of miscarriage and of cardiac malformation and gastroschisis after the use of a prostaglandin synthesis inhibitor in early pregnancy. The risk is believed to increase with dose and duration of therapy. In animals, the administration of a prostaglandin synthesis inhibitor has been shown to result in increased pre- and post-implantation losses and embryo/fetal lethality. In addition, increased incidences of various malformations, including cardiovascular, have been reported in animals given a prostaglandin synthesis inhibitor during the organogenetic period.

From the 20th week of pregnancy onward, ibuprofen use may cause oligohydramnios resulting from foetal renal dysfunction. This may occur shortly after treatment initiation and is usually reversible upon discontinuation. In addition, there have been reports of ductus arteriosus constriction following treatment in the second trimester, most of which resolved after treatment cessation. Therefore, during the first and second trimester of pregnancy, ibuprofen should not be given unless clearly necessary. If ibuprofen is used by a woman attempting to conceive, or during the first or second trimester of pregnancy, the dose should be kept as low and duration of treatment as short as possible. Antenatal monitoring for oligohydramnios and ductus arteriosus constriction should be considered after exposure to ibuprofen for several days from gestational week 20 onward. Ibuprofen should be discontinued if oligohydramnios or ductus arteriosus constriction are found.

During the third trimester of pregnancy, all prostaglandin synthesis inhibitors may expose the fetus to the following:

- Cardiopulmonary toxicity (premature constriction/closure of the ductus arteriosus and pulmonary hypertension)
- Renal dysfunction, which may progress to renal failure with oligohydramnios.

At the end of pregnancy, prostaglandin synthesis inhibitors may expose the mother and the neonate to the following:

- Possible prolongation of bleeding time, an anti-aggregating effect which may occur even at very low doses
 - Inhibition of uterine contractions, which may result in delayed or prolonged labor.
- Consequently, ibuprofen is contraindicated during the third trimester of pregnancy.

Labor and Delivery

Administration of ibuprofen is not recommended during labor and delivery.

The onset of labor may be delayed and the duration increased with a greater bleeding tendency in both mother and child.

Breastfeeding

In the limited studies so far available, ibuprofen appears in the breast milk in very low concentrations. Ibuprofen is not recommended for use in nursing mothers.

EFFECTS ON ABILITY TO DRIVE AND TO USE MACHINES

Following treatment with ibuprofen, the reaction time of patients may be affected. This should be taken into account where increased vigilance is required, e.g., when driving a car or operating machinery. This applies to a greater extent in combination with alcohol.

ADVERSE REACTIONS

The pattern of adverse events reported for ibuprofen is similar to that for other NSAIDs.

- **Gastrointestinal Disorders**

The most commonly observed adverse events are gastrointestinal in nature. Nausea, vomiting, diarrhea, flatulence, constipation, dyspepsia, abdominal pain, melena, hematemesis, ulcerative stomatitis, gastrointestinal hemorrhage and exacerbation of colitis and Crohn's disease (see section Contraindications) have been reported following ibuprofen administration. Less frequently, gastritis, duodenal ulcer, gastric ulcer and gastrointestinal perforation have been observed.

A transient sensation of burning in the mouth or throat may occur with ibuprofen suspension.

- **Immune System Disorders**

Hypersensitivity reactions have been reported following treatment with ibuprofen. These may consist of (a) non-specific allergic reaction and anaphylaxis, (b) respiratory tract reactivity comprising asthma, aggravated asthma, bronchospasm or dyspnea, or (c) assorted skin disorders, including rashes of various types, pruritus, urticaria, purpura, angioedema and, very rarely, erythema multiforme, bullous dermatoses (including Stevens-Johnson syndrome and toxic epidermal necrolysis).

- **Infections and infestations**

Exacerbation of skin infection-related inflammations (e.g., development of necrotizing fasciitis) coinciding with the use of NSAIDs has been described. If signs of an infection occur or get worse during use of Ibuprofen the patient is therefore recommended to go to a doctor without delay.

- **Skin and subcutaneous tissue disorders**

In exceptional cases, severe skin infections and soft-tissue complications may occur during a varicella infection (see also "Infections and infestations").

- **Cardiac and vascular disorders**

Clinical studies suggest that use of ibuprofen, particularly at a high dose (2400mg/ day) may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke) (see section Warnings and Precautions).

The following adverse reactions possibly related to ibuprofen are displayed by MedDRA frequency convention and system organ classification. Frequency groupings are classified according to the subsequent conventions: very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), Very rare ($< 1/10,000$) and Not known (cannot be estimated from the available data).

System organ class	Frequency	Adverse reaction
Infections and infestations	Uncommon	Rhinitis
	Rare	Meningitis aseptic (see section Warnings and Precautions)
Blood and lymphatic system disorders	Rare	Leukopenia, thrombocytopenia, neutropenia, agranulocytosis, aplastic anemia and hemolytic anemia
Immune system disorders	Uncommon	Hypersensitivity
	Rare	Anaphylactic reaction
Psychiatric disorders	Uncommon	Insomnia, anxiety
	Rare	Depression, confusional state

Nervous system disorders	Common	Headache, dizziness
	Uncommon	Paraesthesia, somnolence
	Rare	Optic neuritis
Eye disorders	Uncommon	Visual impairment
	Rare	Toxic optic neuropathy
Ear and labyrinth disorders	Uncommon	Hearing impaired, tinnitus, vertigo
Respiratory, thoracic and mediastinal disorders	Uncommon	Asthma, bronchospasm, dyspnea
Gastrointestinal disorders	Common	Dyspepsia, diarrhea, nausea, vomiting, abdominal pain, flatulence, constipation, melena, hematemesis, gastrointestinal hemorrhage
	Uncommon	Gastritis, duodenal ulcer, gastric ulcer, mouth ulceration, gastrointestinal perforation
	Very rare	Pancreatitis
	Not known	Colitis and Crohn's disease
Hepatobiliary disorders	Uncommon	Hepatitis, jaundice, hepatic function abnormal
	Very rare	Hepatic failure
Skin and subcutaneous tissue disorders	Common	Rash
	Uncommon	Urticaria, pruritus, purpura, angioedema,
	Very rare	Severe cutaneous adverse reactions (SCARs) including Erythema multiforme, exfoliative dermatitis, bullous reactions Including Stevens-Johnson syndrome, and toxic epidermal necrolysis.)
	Not Known	Drug reaction with eosinophilia and systemic symptoms (DRESS syndrome), Acute generalised exanthematous pustulosis (AGEP), photosensitivity reactions
Metabolism and Nutrition Disorders	Not Known	Decreased Appetite Hypokalaemia*
Renal and urinary disorders	Uncommon	Nephrotoxicity in various forms, e.g., tubulointerstitial nephritis, nephrotic syndrome and renal failure
	Very rare	Acute renal failure

	Not Known	Ureteric colic, dysuria Renal tubular acidosis*
General disorders and administration site conditions	Common	Fatigue
	Rare	Edema
Cardiac disorders	Very rare	Cardiac failure, myocardial infarction (also see section Warnings and Precautions)
	Not Known	Kounis Syndrome
Vascular disorders	Very rare	Hypertension

*Renal tubular acidosis and hypokalaemia have been reported in the post-marketing setting typically following prolonged use of the ibuprofen component at higher than recommended doses.

OVERDOSAGE

- Toxicity**

Signs and symptoms of toxicity have generally not been observed at doses below 100 mg/kg in children or adults. However, supportive care may be needed in some cases. Children have been observed to manifest signs and symptoms of toxicity after ingestion of 400 mg/kg or greater.

- Symptoms**

Most patients who have ingested significant amounts of ibuprofen will manifest symptoms within 4 to 6 hours.

The most frequently reported symptoms of overdose include nausea, vomiting, abdominal pain, lethargy and drowsiness. Central nervous system (CNS) effects include headache, tinnitus, dizziness, convulsion, and loss of consciousness. Nystagmus, metabolic acidosis, hypothermia, renal effects, gastrointestinal bleeding, coma, apnea and depression of the CNS and respiratory system have also been rarely reported. Cardiovascular toxicity, including hypotension, bradycardia and tachycardia, has been reported. In cases of significant overdose, renal failure and liver damage are possible. Large overdoses are generally well tolerated when no other drugs are being taken. In serious poisoning metabolic acidosis may occur and the prothrombin time/INR may be prolonged, probably due to interference with the actions of circulating clotting factors.

Prolonged use at higher than recommended doses may result in severe hypokalaemia and renal tubular acidosis. Symptoms may include reduced level of consciousness and generalised weakness (see section Warnings and Precautions and section Adverse Reactions).

- Treatment**

There is no specific antidote for ibuprofen overdose. Patients should be treated symptomatically as required. Within one hour of ingestion of a potentially toxic amount, activated charcoal should be considered. If necessary, serum electrolyte balance should be corrected. For the most current information, contact the local poison control center.

PHARMACOLOGIC PROPERTIES

Pharmacodynamic Properties

Ibuprofen is a propionic acid derivative with analgesic, anti-inflammatory and antipyretic activity. The drug's therapeutic effects are thought to result from its inhibitory effect on the enzyme cyclooxygenase, which results in a marked reduction in prostaglandin synthesis.

Experimental data suggest that ibuprofen may inhibit the effect of low dose aspirin on platelet aggregation when they are dosed concomitantly. In one study, when a single dose of ibuprofen 400 mg was taken within 8 hours before or within 30 minutes after immediate release aspirin dosing (81 mg), a decreased effect of acetylsalicylic acid on the formation of thromboxane or platelet aggregation occurred. However, the limitations of these data and the uncertainties regarding extrapolation of ex vivo data to the clinical situation imply that no firm conclusions can be made for regular ibuprofen use, and no clinically relevant effect is considered to be likely for occasional ibuprofen use.

Pharmacokinetic Properties

Ibuprofen is rapidly absorbed from the gastrointestinal tract, peak serum concentrations occurring one to two hours after administration. The elimination half-life is approximately two hours. Ibuprofen is metabolized in the liver to two inactive metabolites and the kidney excretes these, together with unchanged ibuprofen, either as such or as conjugates. Excretion by the kidney is both rapid and complete. Ibuprofen is extensively bound to plasma proteins.

STORAGE

The storage statement should be established in accordance with relevant national/regional requirements.

PACK SIZE

Brufen® 100mg/5ml Suspension (90 ml)

Brufen® 100mg/5ml Suspension (60 ml)

STORAGE

Store below 30°C

SHELF LIFE

36 months

MANUFACTURER

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